FAQs

What is hydrogen water?

Hydrogen water or hydrogen-rich water (also called hydrogen-enriched water) simply means normal **water** (H_2O) that contains dissolved **hydrogen gas (H_2**). For example, there are carbonated waters or beverages (soda pop), which contain dissolved carbon dioxide gas (CO_2), or oxygen water, which contains dissolved oxygen gas (O_2). Similarly, you can have water that contains dissolved hydrogen gas.

Think of it this way: you can make hydrogen water by taking a tank of hydrogen gas (just like tanks of helium used to fill balloons or tanks of oxygen gas used in hospitals), and bubbling it into a glass of water. There are also many **other methods** to make hydrogen water, but this may help you better understand what hydrogen water is. It is simply water that contains **dissolved** hydrogen gas.

Doesn't water already have hydrogen in it because water is H2O?

The water molecule has two hydrogen atoms, chemically bound to the oxygen atom. This is different from the hydrogen gas molecule (H2), which is just two hydrogen atoms bound only to each other. Here's an example: we need oxygen (O2) to live, so why can't we just get our oxygen from drinking water, H2O? It's because the oxygen is chemically tied up in the water molecule. We need available oxygen gas, (O2) that is not bound to other atoms or molecules. In the same way, in order for the dissolved hydrogen gas (H2) to benefit us, it must be in an unbound form, and therefore available for therapeutic benefit. This is why water is not explosive or doesn't burn. Although, it contains hydrogen, which is flammable, and oxygen, which fire needs to burn, the hydrogen and oxygen are bonded together to form water (H2O). Thus, water is not flammable-in fact, H2O is what we use to extinguish fires. Furthermore, virtually everything has hydrogen atoms in it, but those hydrogen atoms are chemically tied up with other things. For example, a water molecule has two hydrogen atoms that are chemically tied up with the oxygen. Or, a sugar molecule like glucose contains 12 hydrogens, but those hydrogens are all bound to other carbon and oxygen atoms. In hydrogen water, the hydrogen that is shown to be therapeutic is the available dissolved hydrogen in its diatomic form, called molecular hydrogen.

How to get my Daily Dose of Hydrogen?

There are several effective delivery methods. Hydrogen can be administered via direct inhalation of the gas. However, the simplest method is to infuse the therapeutic gas into water, like making carbonated water. Many studies have used hydrogen water to demonstrate the therapeutic efficacy of the gas as it presents the most comfortable to administer and can be as simple as drinking 1-3 liters of water per day.

When is the best time to drink the water?

We have suggested protocols of drinking the water, which will depend on your current health condition and your desired outcomes. The amount of water and concentration you consume will vary depending on whether you are looking for general wellness, sports recovery or to support a metabolic illness. Some general recommendations are to consume at least (1) 12 oz glass every morning upon waking. Drink (30) minutes before or after meals. Consume water before and after workouts and stop drinking 1-2 hours before you go to bed. If it is hot outside or you are transpiring or perspiring more, drink more. If you are consuming diuretics like coffee or alcohol, consume (1) glass of hydrogen rich water for each cup you consume to counter the fluids that will be lost. Here is the best rule to follow, any time you feel a bit fatigued during the day, take a water break and rehydrate with fresh hydrogen water!

I thought that if water is "hydrogen rich", then it must be acidic?

Great question! If the water is rich in **positive hydrogen ions (H+),** then yes, it IS acidic. But in this case, we're talking about neutral hydrogen gas (H2), which is two hydrogen atoms tied together. It can be confusing to hear "hydrogen water" because we usually think of hydrogen (meaning the hydrogen ion, H+) as acidic, and that is basically the definition of pH. The p stands for potential or power, meaning a mathematical exponent (in this case a logarithmic function), and the H stands for the hydrogen ion, which is just a proton and no electron. So pH literally means the logarithmic concentration of the hydrogen ion. But when we say "hydrogen water" we are referring to dihydrogen or molecular hydrogen, which is a neutral gas that is dissolved in the water.

I read that if you add hydrogen to water, then it makes hydrogen peroxide?

Water has the chemical formula H_2O, and hydrogen peroxide has the chemical formula H_2O_2, which by comparison contains an extra oxygen, not hydrogen. So it does not, indeed it cannot, form hydrogen peroxide. The fact is, hydrogen gas does not bond to or react with the water molecules, it just dissolves into the water. It does not create some novel molecule like H_4O, which would in fact be chemically impossible to form. Therefore, hydrogen water and hydrogen peroxide are completely different substances. Furthermore, hydrogen peroxide cannot be used to generate H_2 gas or make hydrogen-rich water.

Won't any dissolved hydrogen gas immediately escape out of the water?

Yes, it does immediately start coming out of the water, but it doesn't just vanish immediately. Depending on the surface area, agitation, etc., the hydrogen gas can stay in the water for a few hours or longer before it drops below a therapeutic level. This is much like carbonated water or soda that contains carbon dioxide gas (CO2), but because it does leave, it is best to drink the water promptly before it goes "flat".

How much hydrogen water should I drink to get the benefits?

That is the same question scientists are asking and is still under investigation. However, the animal and human studies generally provide about 0.5 to 1.6 mg or more of H2 per day, and these doses show statistically significant benefits. So, if your water has a concentration of 1 mg/L (equivalent to 1 ppm, parts per million), then **two liters** will give you 2 mg of H2. Although the effective concentration for some people and some diseases may be lower and/or higher, these doses are simply what have been seen to exert benefits. The key is to consume hydrogen rich water consistently so the effects can compound over time.

Isn't hydrogen gas explosive?

Yes, it is VERY explosive. Hydrogen is the most energy-dense molecule by mass. But, when the gas is dissolved in water it is not explosive at all, just like if you mixed gunpowder in water it wouldn't be explosive either. Even when it is in the air, it is only flammable above a 4.6% concentration by volume, which is not a concern when talking about hydrogen-rich water.

Is your hydrogen machine considered a medical device?

In the USA, no hydrogen machine is approved as a medical device. Because Hydrogen is a natural substance and cannot be patented, you will probably never see a hydrogen machine in USA based hospitals. There are though, many wellness centers, sports recovery and antiaging clinics that do incorporate Molecular Hydrogen into their health & wellness protocols.
There are though, many hospitals in Japan, Korea, China and other parts of Asia that do have approved hydrogen machines as medical devices.

Is hydrogen safe?

Yes. Molecular hydrogen is considered as GRAS (generally regarded as safe) by the FDA. In fact, hydrogen gas is even safe at concentrations hundreds of times higher than what is being used for therapy. Here are a few examples: Hydrogen's safety was first shown in the late 1800s, where hydrogen gas was used to locate gunshot wounds in the intestines. The reports showed that there were never any toxic effects or irritation to even the most sensitive tissues. Another good example of its safety is that hydrogen gas has been used in deep sea diving since 1943 (at very high concentrations) to prevent decompression sickness. Studies have shown no toxic effects from hydrogen when at very high levels and pressures of 98.87% H_2 and 1.26% O_2 at 19.1 atm. Furthermore, hydrogen gas is natural to the body because after a fiber-rich meal, our gut bacteria can produce liters of hydrogen daily (which is yet another benefit from eating fruits and vegetables). In short, hydrogen gas is very natural to our bodies, not like a foreign or alien substance that can only be synthesized in a chemistry lab.

When was hydrogen's therapeutic benefits first discovered?

The earliest account of hydrogen gas having medicinal properties was in 1798, for things like inflammation. But it didn't become a popular topic among scientists until 2007, when an article about the benefits of hydrogen was published in the prestigious journal of Nature Medicine by Dr. Ohta's group. Since that time there have been thousands of studies published from 44 countries around the word and growing. The best study will be will you start drinking the water on a consistent daily basis.